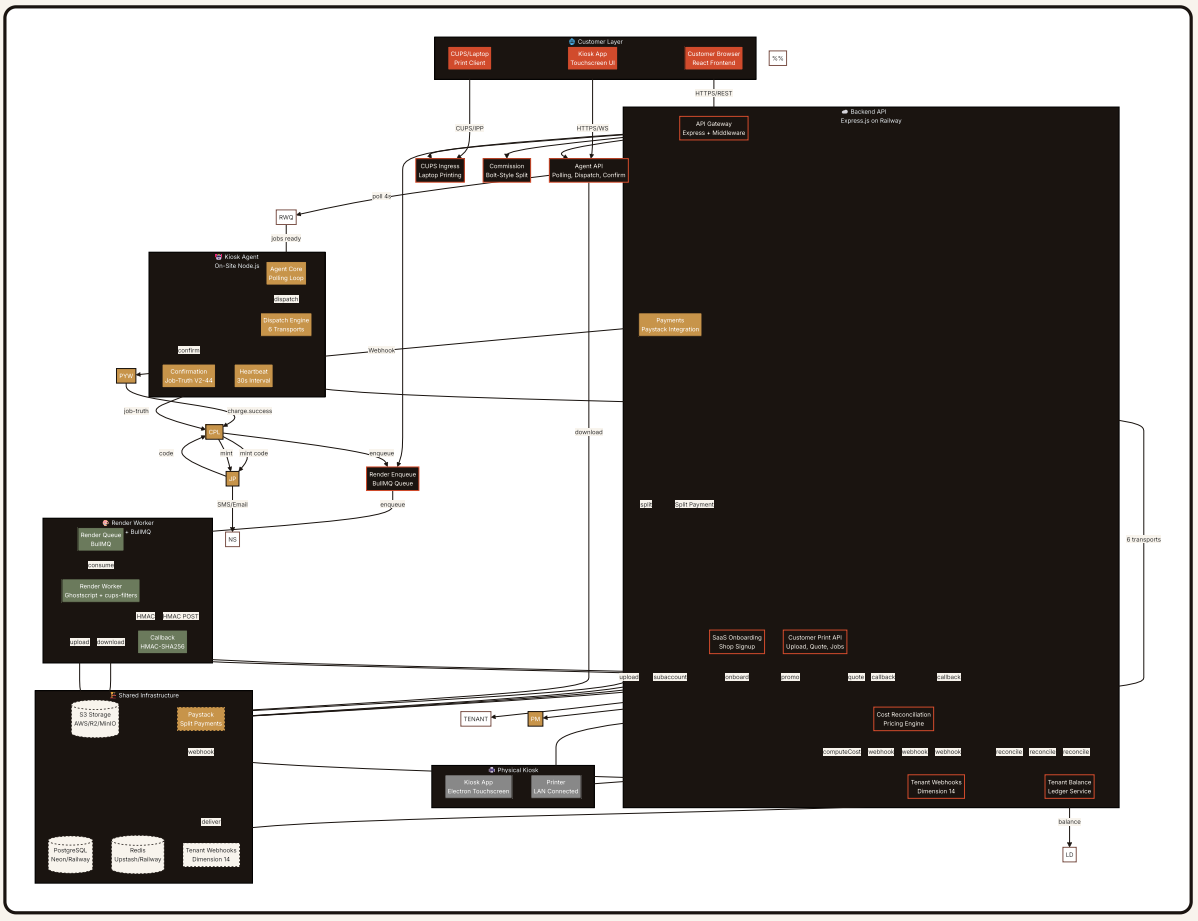
PrintLoop SaaS v2

Complete Architectural Flow Mapping

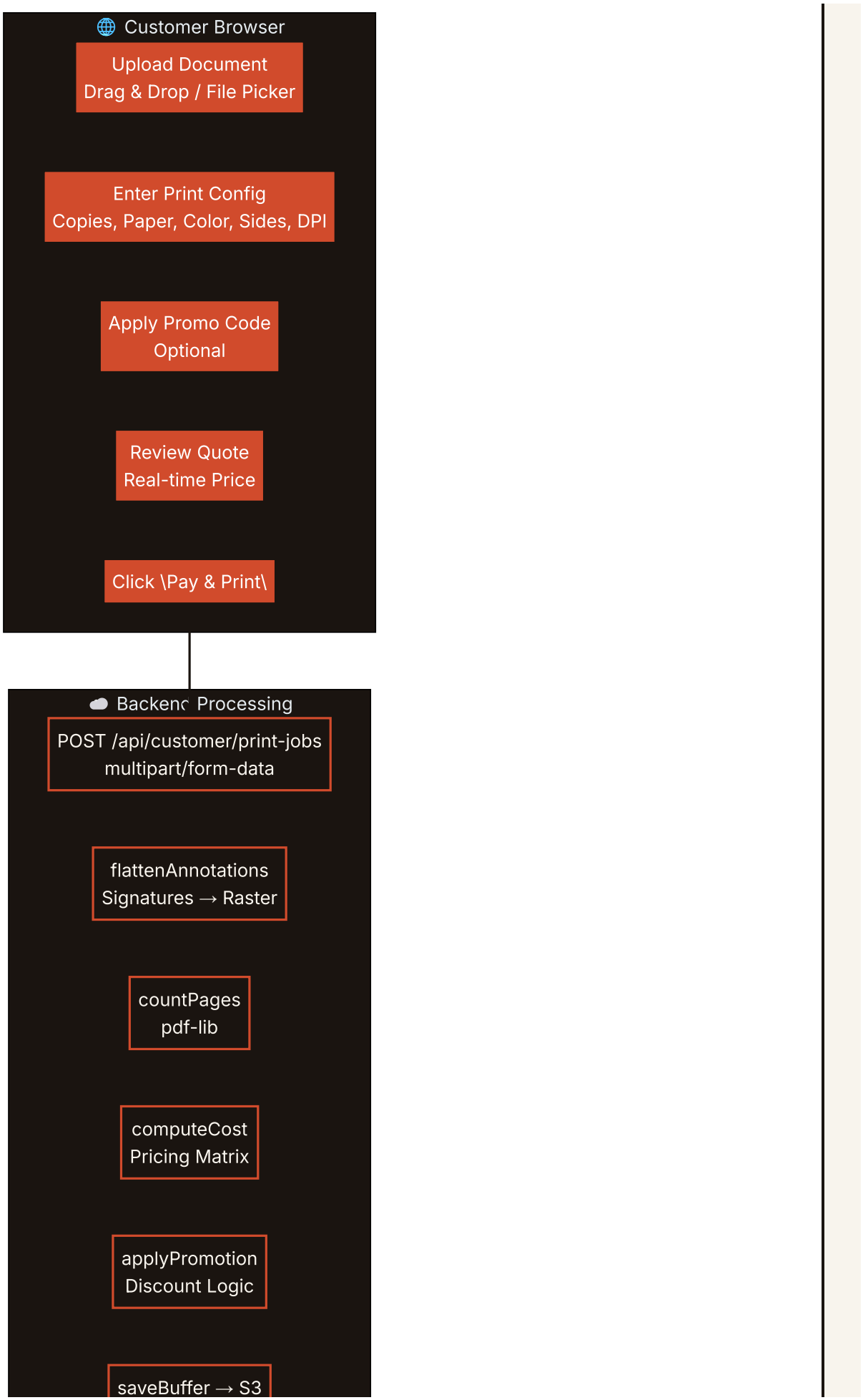
v2.0 Architecture Flow Mapping

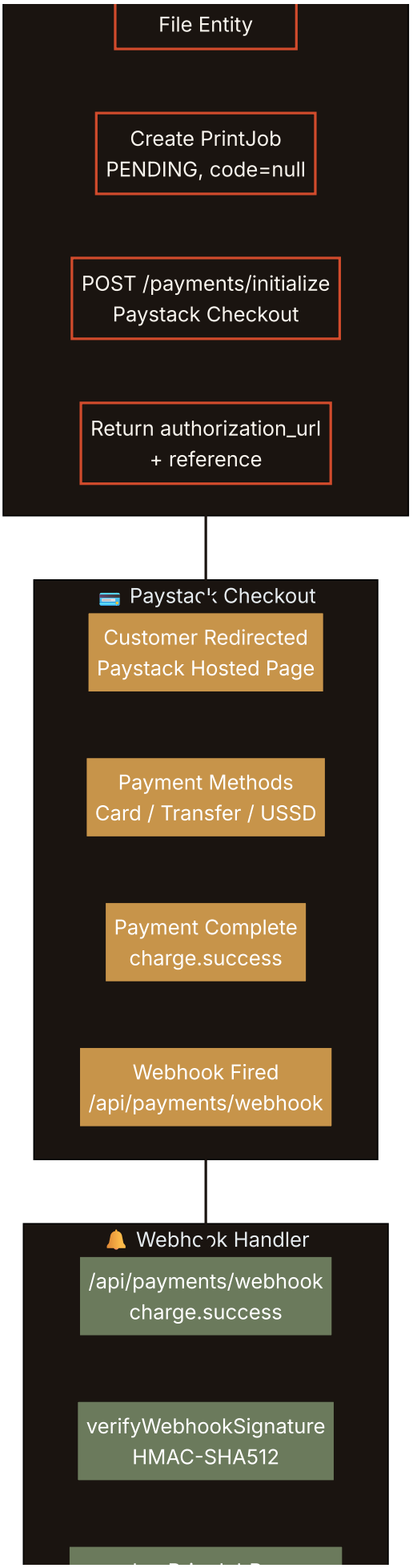
☐ Customer Actions ☐ System Processes ☐ Payment/Money ☐ Rendering/Printing

System Overview



1. Customer Upload Flow


Customer Browser
Upload Document
Drag & Drop / File Picker
Enter Print Config
Copies, Paper, Color, Sides, DPI
Apply Promo Code
Optional
Review Quote
Real-time Price
Click \Pay & Print\
Backend Processing
POST /api/customer/print-jobs
multipart/form-data
flattenAnnotations
Signatures → Raster
countPages
pdf-lib
computeCost
Pricing Matrix
applyPromotion
Discount Logic
saveBuffer → S3


File Entity
Create PrintJob
PENDING, code=null
POST /payments/initialize
Paystack Checkout
Return authorization_url
+ reference
Paystack Checkout
Customer Redirected
Paystack Hosted Page
Payment Methods
Card / Transfer / USSD
Payment Complete
charge.success
Webhook Fired
/api/payments/webhook
Webhook Handler
/api/payments/webhook
charge.success
verifyWebhookSignature
HMAC-SHA512



2. Cloud Render → Kiosk Spool Pipeline (V2-6)



3. Kiosk Agent Flow (On-Site)



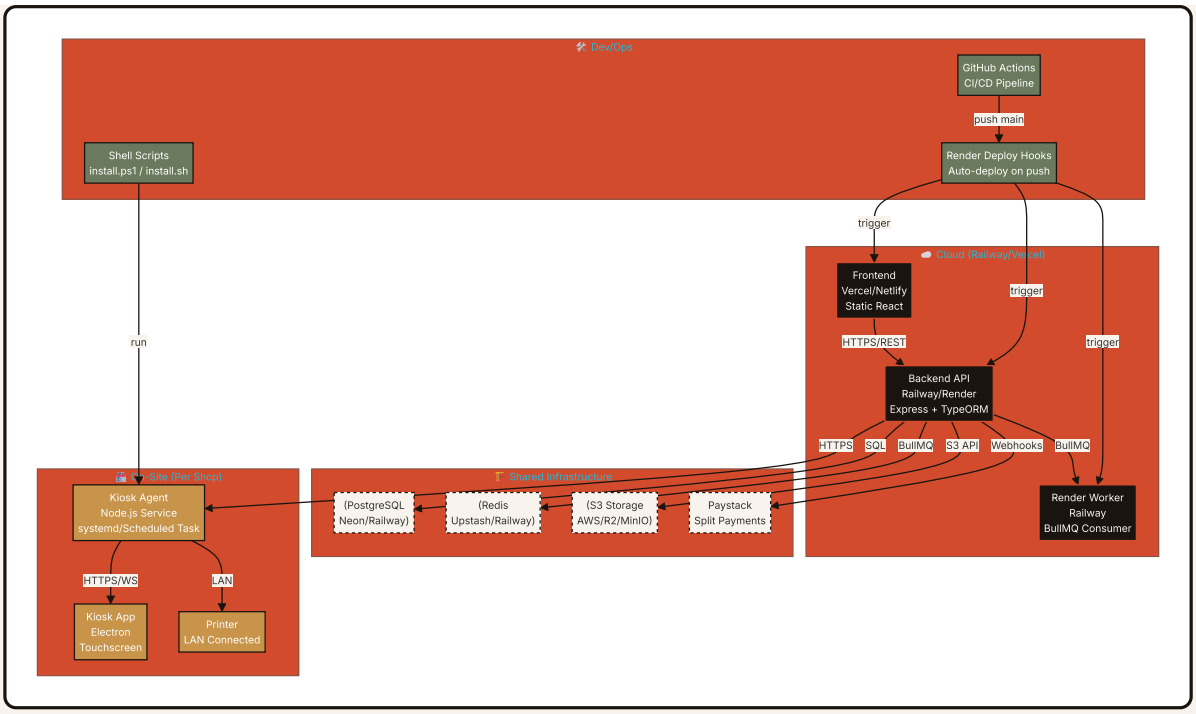
4. Payment Flow (Paystack)



5. Job Lifecycle State Machine



6. Deployment Topology



7. Entity Relationship Diagram



Syntax error in text

mermaid version 10.9.8

8. Sequence: Customer Upload to Print



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mermaid version 10.9.8

9. Security Flow



10. API Flow Diagram

